

# RSTS Analysis GUI User Guide (Version 9.8)

## Overview

The RSTS (Raman and Thomson Scattering) Analysis GUI is a comprehensive tool for analyzing plasma diagnostics data from combined Raman and Thomson scattering experiments. It provides real-time visualization, multiple fitting algorithms, and calibration tools for extracting electron temperature ( $T_e$ ) and electron density ( $n_e$ ) from spectroscopic measurements.

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## Main Display Window

The GUI consists of four main plotting areas arranged in a 2×2 grid:

### Top Row Plots

#### 1. Raman Image (Top Left)

- Displays the processed Raman scattering image with background subtraction applied
- **Yellow dashed rectangle:** Region of Interest (ROI) showing the vertical integration window
- **Red shaded rectangle:** Excluded pixel columns (typically around the laser line at 532 nm)
- Color scale indicates signal intensity

#### 2. Thomson Image (Top Right)

- Shows the processed Thomson scattering image with background subtraction
- Uses median filtering for noise reduction
- Same ROI and exclusion zone markers as Raman image
- Brighter regions indicate stronger scattering signal

### Bottom Row Plots

#### 3. Raman Spectrum (Bottom Left)

- **Blue solid line:** Experimental Raman spectrum (integrated from ROI)
- **Red dashed line:** Theoretical model spectrum (scaled for comparison)
- Title displays the calibration factor used to convert counts to physical units
- X-axis: Wavelength (nm)
- Y-axis: Intensity (counts)

#### 4. Thomson Spectrum (Bottom Right)

- **Blue line/points:** Experimental Thomson scattering data
- **Red line:** Initial Gaussian fit (incoherent approximation)
- **Green line:** Advanced coherent fit (when Shape/Intensity calibration is performed)
- Title shows fit quality (SSR,  $R^2$ ) and extracted plasma parameters ( $T_e$ ,  $n_e$ )
- X-axis: Wavelength (nm)
- Y-axis: Intensity (counts)

#### Save Buttons

Below each plot are save buttons that export the current view as JPG files to the same directory as your data files. Filenames automatically include relevant parameters ( $T_e$ ,  $n_e$ , calibration factor, fit type, SSR).

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### Analysis Controls Panel

#### Region of Interest (ROI) Parameters

##### ROI yc (Row 1, Column 1)

- Center row pixel for vertical integration
- Default: Auto-centered based on image dimensions
- Typical range: 20-700 depending on detector

##### ROI wy (Row 2, Column 1)

- Width (height) of ROI in pixels for vertical integration
- Determines how many rows are averaged to create the spectrum
- Typical values: 20-60 pixels
- Larger values increase signal but may include more noise

#### Exclusion Zone Parameters

##### Excl Start (Row 1, Column 3) Excl End (Row 2, Column 3)

- Define the pixel column range to exclude from fitting
- Used to block the intense laser line and Rayleigh scattering
- Typically spans 15-70 pixels centered on the laser wavelength
- Red shaded area in image plots shows this region

## Laser and Detection Parameters

### Pressure (Pa) (Row 2, Column 7)

- Gas pressure for Raman calibration
- Standard atmospheric: 101325 Pa
- Affects the theoretical Raman spectrum calculation

### Raman Shots (Row 3, Column 1) Thomson Shots (Row 4, Column 1)

- Number of laser shots accumulated in each measurement
- Used for energy normalization in calibration factor calculation

### Raman E (mJ) (Row 3, Column 3) Thomson E (mJ) (Row 4, Column 3)

- Laser pulse energy in millijoules per shot
- Critical for absolute intensity calibration

## Wavelength Calibration Coefficients

### a coeff (Row 3, Column 5)

- Linear wavelength calibration slope:  $\lambda(\text{nm}) = a \times \text{pixel} + b$
- Typical values: 0.005-0.03 nm/pixel depending on grating and camera

### b coeff (Row 3, Column 7)

- Wavelength calibration offset in nm
- Represents wavelength at pixel 0
- Typical values: 500-530 nm for 532 nm laser systems

### Linewidth (nm) (Row 4, Column 5)

- Instrument broadening (FWHM) of the spectrometer
- Accounts for finite spectral resolution
- Typical values: 0.05-0.7 nm depending on grating groove density
- Used in both Raman model and Thomson deconvolution

## Plasma Parameters

### Te (eV) (Row 5, Column 1)

- Electron temperature in electron volts

- Editable field that can be manually adjusted
- Automatically updated after Gaussian fit or Shape/Intensity calibration
- Typical range: 0.5-15 eV for most plasmas

#### **$\alpha$ (alpha)** (Row 5, Column 3)

- Scattering parameter  $\alpha = 1/(k \cdot \lambda_D)$  where  $\lambda_D$  is Debye length
- **Non-editable** (calculated from  $T_e$  and  $n_e$ )
- $\alpha < 1$ : Incoherent scattering regime (Gaussian approximation valid)
- $\alpha > 1$ : Coherent scattering regime (requires full form factor)

#### **$n_e$ (m<sup>-3</sup>)** (Row 5, Column 5)

- Electron density in particles per cubic meter
- Editable field for manual adjustment
- Automatically updated after fitting
- Typical range:  $10^{18}$ - $10^{22}$  m<sup>-3</sup>

#### **$\gamma$ (gamma)** (Row 5, Column 7)

- Duty cycle or temporal fraction parameter
- Accounts for time-varying plasma density
- Default: 1.0 (continuous plasma)
- Values < 1.0 indicate pulsed or intermittent plasma
- **Editable** as of v9.8

### **Gas Composition Parameters**

#### **Temperature (K)** (Row 6, Column 1)

- Gas temperature for Raman spectrum calculation
- Affects rotational-vibrational line intensities
- Typical range: 273-400 K
- Default: 293.15 K (room temperature)

#### **N<sub>2</sub> fraction** (Row 6, Column 3) **O<sub>2</sub> fraction** (Row 6, Column 5)

- Relative composition of nitrogen and oxygen
- Must sum to 1.0 (automatically normalized if not)

- Default air composition:  $N_2=0.79$ ,  $O_2=0.21$
- For pure gases: set one to 1.0 and the other to 0.0

## Multi-start Optimization

### Multi-starts (Row 7, Column 1)

- Number of random initializations for optimization algorithms
  - Higher values increase robustness but take longer
  - Default: 6
  - Range: 1-20
  - Used by "Recalibrate from Raman", "Shape Calibration", and "Intensity Calibration"
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## Main Analysis Buttons

### Row 8 Buttons (Primary Fitting Functions)

#### Recalculate TS (Columns 1-2)

- Regenerates Thomson spectrum using current manual  $T_e$ ,  $n_e$ ,  $\alpha$ , and  $\gamma$  values
- Uses coherent scattering form factor with user-specified parameters
- Does **not** perform any fitting—only forward calculation
- Updates the green "Recalculated" curve in the TS plot
- Useful for exploring parameter space manually

#### Shape Calibration (Columns 3-4)

- Performs multi-start coherent Thomson scattering fit
- Optimizes  $T_e$  and  $\alpha$  by fitting the spectral **shape only**
- Does **not** use absolute intensity—normalizes both data and model
- Returns "instantaneous"  $n_e$  ( $n_{e\_inst}$ ) derived from fitted  $\alpha$  and  $T_e$
- Best for scenarios where calibration factor is uncertain
- Updates  $T_e$ ,  $\alpha$ , and  $n_e$  fields with results
- Shows green "Shape Fit" curve with SSR and  $R^2$  statistics

#### Incoherent TS Fit (Columns 5-6, green background)

- Re-runs the complete analysis pipeline with current GUI parameters

- Performs Gaussian fit (incoherent approximation,  $\alpha \ll 1$ )
- Includes wing-based deconvolution to correct for Raman contamination
- Updates all four plots and resets fit to red Gaussian curve
- Use this after changing ROI, exclusion zones, or calibration coefficients
- **This is the default initial fit shown on startup**

**Intensity Calibration** (Columns 7-9, blue background)

- Two-stage advanced coherent fitting algorithm:
  - **Stage 1:** Fit  $T_e$  and  $\alpha$  from spectral shape (like Shape Calibration)
  - **Stage 2:** Calculate  $n_e$  from total integrated intensity using calibration factor
- Provides most accurate results when calibration factor is well-known
- Requires good Raman calibration
- Returns  $T_e$  with error bars ( $\pm \Delta T_e$ ) and  $n_e$  with error bars ( $\pm \Delta n_e$ )
- Shows green "Intensity Cal" curve
- **Recommended for final analysis**

**Key Differences: Shape vs. Intensity Fit**

| Feature            | Shape Calibration                                     | Intensity Calibration                        |
|--------------------|-------------------------------------------------------|----------------------------------------------|
| What it fits       | Spectral shape only (normalized)                      | Shape + absolute intensity                   |
| Parameters         | $T_e$ , $\alpha$ (amplitude free)                     | $T_e$ , $\alpha$ , then $n_e$ from intensity |
| Uses calib factor? | No                                                    | Yes (critical)                               |
| Best for           | Unknown calibration, relative measurements            | Absolute density determination               |
| $n_e$ result       | Instantaneous ( $n_{e\_inst}$ from $\alpha$ , $T_e$ ) | Time-averaged (from total signal)            |
| Sensitivity        | Robust to calibration errors                          | Requires accurate Raman calibration          |
| Error bars         | Yes (from fit covariance)                             | Yes (propagated through both stages)         |

**When to use each:**

- **Shape Calibration:** When you don't trust your calibration factor, or want to verify  $\alpha$  regime
- **Intensity Calibration:** When you have good Raman data and want the most accurate absolute  $n_e$

**Secondary Control Buttons**

**Recalibrate from Raman** (Row 4, Columns 7-8)

- Advanced multi-stage optimization of Raman calibration:
  1. **Stage 0:** Intelligent peak-matching algorithm aligns experimental and theoretical peak positions
  2. **Stage 1:** Multi-start optimization of calibration factor, linewidth, and gas temperature
- Updates a, b coefficients and linewidth
- Re-analyzes entire dataset with improved calibration
- Use when peaks are misaligned or calibration factor seems wrong
- **Computationally intensive** (may take 10-30 seconds)

#### **Recalc Raman Spectrum** (Row 6, Columns 7-9, tan background)

- Recalculates theoretical Raman model using current gas composition and temperature
- Updates only the Raman spectrum plot (bottom left)
- Does **not** reprocess images or refit Thomson data
- Use after changing N<sub>2</sub>/O<sub>2</sub> fractions, temperature, or linewidth

#### **Toggle TS plot style** (Row 2, Columns 5-6)

- Switches Thomson data display between:
  - **Line plot:** Continuous blue line (default)
  - **Marker plot:** Individual blue circles (useful for noisy data)
- Does not affect fitting, only visualization

#### **Fit Adjustment Controls** (Row 9)

< **button** (Column 6) > **button** (Column 7) **Shift: 0 px** (Column 8)

- Manually shift the fitted curve left (<) or right (>) by 1 pixel per click
- Useful for correcting small wavelength calibration errors
- Shift value is displayed and persists until reset
- Does **not** modify underlying calibration—only visual/analysis alignment

## **Utilities Panel**

### **Analysis Tools**

#### **Decouple Plots**

- Exports all four plots to separate, independent figure windows

- Useful for saving high-resolution images or detailed examination
- Plots remain interactive in new windows

### **Plot EEDF**

- Generates Electron Energy Distribution Function plot
- Converts wavelength-shifted spectrum to electron energy space
- Shows blue wing (short wavelength) and red wing (long wavelength) separately
- Uses logarithmic y-axis to reveal distribution shape
- Useful for checking for non-Maxwellian features

### **Save TS Data to CSV**

- Exports Thomson spectrum data (experimental + fitted) to CSV file
- Filename includes fit type,  $T_e$ ,  $n_e$ ,  $\alpha$ , and SSR
- Three columns: Wavelength\_nm, Experimental\_Intensity, Fitted\_Intensity
- Saved to same directory as Thomson image file

## **File Management**

### **Raman File**

- Displays current Raman image path
- **Load Img:** Select new Raman image file (.tif, .tiff, .png)
- **Load BG:** Select Raman background image for subtraction

### **Raman BG**

- Shows loaded background file path or "No file loaded"

### **Thomson File**

- Displays current Thomson image path
- **Load Img:** Select new Thomson image file
- **Load BG:** Select Thomson background image for subtraction

### **Thomson BG**

- Shows loaded background file path or "No file loaded"



## Thomson Files Browser

### Files in current folder

- Listbox showing all image files in the Thomson image directory
- Sorted by date (most recent first)
- Current file is highlighted
- Click any file to instantly switch to it and reanalyze
- Useful for batch processing or comparing sequential shots

### Refresh File List

- Updates the file list to show newly added files
  - Use after adding new data to the folder
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## Status Indicator

**Status: Ready** (Bottom left of controls)

- Displays current operation status
  - **Green text:** Ready, operations complete
  - **Blue text:** Processing (recalculating, fitting, loading)
  - **Red text:** Error condition
  - Updates automatically during all operations
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## Typical Workflow

### 1. Initial Setup

1. Launch GUI—loads default dataset (configured in code)
2. Check image displays—verify ROI (yellow box) captures signal region
3. Check exclusion zone (red box) blocks laser line

### 2. Basic Analysis

1. Click "**Incoherent TS Fit**" to perform Gaussian analysis
2. Examine  $R^2$  value—should be  $> 0.5$  for good data
3. Note  $T_e$  and  $n_e$  values in title and editable fields

### 3. Calibration Refinement (if needed)

1. If Raman peaks misaligned: click **"Recalibrate from Raman"**
2. Adjust Multi-starts to 10-15 for better optimization
3. Wait for convergence (~10-30 seconds)
4. Check that Raman model (red dashed) now matches data (blue)

### 4. Advanced Coherent Fitting

1. Check  $\alpha$  value—if  $\alpha > 0.3$ , coherent effects are important
2. Click **"Shape Calibration"** first to verify  $T_e$  and  $\alpha$
3. Then click **"Intensity Calibration"** for final  $n_e$  determination
4. Compare results with Gaussian fit—coherent fit typically gives slightly different  $T_e$

### 5. Manual Adjustments

1. Edit  $T_e$ ,  $n_e$ , or  $\gamma$  fields manually if needed
2. Click **"Recalculate TS"** to see how curve changes
3. Use  $</>$  buttons to fine-tune wavelength alignment

### 6. Data Export

1. Use **"Save TS Spectrum"** button to save current view as JPG
2. Click **"Save TS Data to CSV"** to export numerical data
3. Use **"Decouple Plots"** to export all plots to separate figures

### 7. Batch Processing

1. Use file browser to select next shot
2. GUI automatically reanalyzes
3. Review results and save
4. Repeat for entire dataset

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## Troubleshooting

### Problem: Gaussian fit looks poor (low $R^2$ )

- Check if  $\alpha > 0.5$  (coherent regime)—use Shape/Intensity calibration instead

- Verify exclusion zone doesn't cut off Thomson wings
- Try expanding ROI wy to improve signal-to-noise

**Problem: Raman peaks don't align with model**

- Click "Recalibrate from Raman" with Multi-starts  $\geq 10$
- Manually adjust a and b coefficients if automated routine fails
- Check that gas composition ( $N_2/O_2$ ) matches actual experiment

**Problem: ne values seem too high/low by factor of 2-10**

- Verify Raman and Thomson shot counts are correct
- Check laser energy values (mJ)
- Ensure background subtraction was properly applied
- Try "Intensity Calibration" instead of Gaussian fit

**Problem: Fit curve doesn't match data shape**

- If  $\alpha > 1$ , must use coherent fitting (Shape/Intensity buttons)
- Check that linewidth is reasonable (0.05-0.7 nm)
- Verify temperature and gas composition for Raman contamination

**Problem: Te changes dramatically between Gaussian and coherent fits**

- Normal for  $\alpha > 0.5$ —coherent form factor is more accurate
- Wing-based deconvolution in Gaussian fit can be sensitive to noise
- Trust coherent fit results in high- $\alpha$  regime

## Technical Notes

### Physics Background

- **Incoherent scattering ( $\alpha \ll 1$ ):** Individual electrons scatter independently, Gaussian profile
- **Coherent scattering ( $\alpha \geq 1$ ):** Collective plasma effects, non-Gaussian profile with wings
- **Form factor:**  $S(k, \omega)$  describes the collective scattering including plasma frequency and Landau damping

### Fit Quality Metrics

- **SSR (Sum of Squared Residuals):** Lower is better, but absolute value depends on signal level

- **R<sup>2</sup> (Coefficient of Determination):** 0 to 1, > 0.7 is good, > 0.9 is excellent
- Always check visually—statistics can be misleading with outliers

## Error Estimation

- Error bars calculated from fit covariance matrix (Jacobian method)
  - Assumes Gaussian noise and proper fitting convergence
  - Intensity calibration propagates errors through both fitting stages
  - Typical uncertainties:  $\Delta T_e/T_e \sim 5\text{-}20\%$ ,  $\Delta n_e/n_e \sim 10\text{-}30\%$
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## Version History

### V9.8 (Current)

- Added automatic  $T_e$ ,  $n_e$ ,  $\alpha$  field updates after Gaussian fit
- Gamma field now editable
- Fixed "Recalculate TS" button functionality
- Added status message display
- Enhanced "Recalibrate from Raman" with full multi-start optimization
- Added multi-starts control for all optimization routines
- Added save buttons for each plot axis

### V9.7

- Fixed missing helper functions for "Optimize All" and "Recalculate TS"
  - Improved error handling
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## Default Datasets

The code includes 4 pre-configured datasets (change with `default_set` flag):

1. **NYU Data:** Air plasma, 532 nm, 2400 gr/mm grating
2. **CC Plasma-Liquid:** Water plasma interaction, 366  $\mu\text{s}$  timing
3. **E-Beam (Texas A&M):** Argon plasma, low pressure (1.46 mTorr)
4. **RF Jet (WUSTL):** Atmospheric pressure RF plasma jet

These serve as examples and testing references.

